FORMULAE LIST

Volume of a sphere: $V = \frac{4}{3}\pi r^3$ Volume of a cone: $V = \frac{1}{3}\pi r^2 h$ Volume of a pyramid: $V = \frac{1}{3}Ah$

1. The mass of the Moon is approximately 7.35×10^{22} kilograms. (2)

   The Earth is approximately 81 times heavier.

   Calculate the approximate mass of the Earth.

   Give your answer in scientific notation.

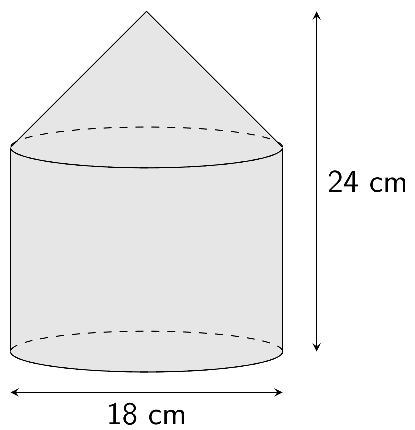
2. The value of a car depreciates by 13% each year. (3)

   The car was valued at £17,400 in 2021.

   Calculate the value of the car in 2026.

3. A child uses two wooden blocks, in the shape of a cone and a cylinder, to make a tower, as below. (5)

   The **cylindrical** block has a height of 17 centimetres.



   Calculate the volume of the tower.

   Give your answer correct to two significant figures.

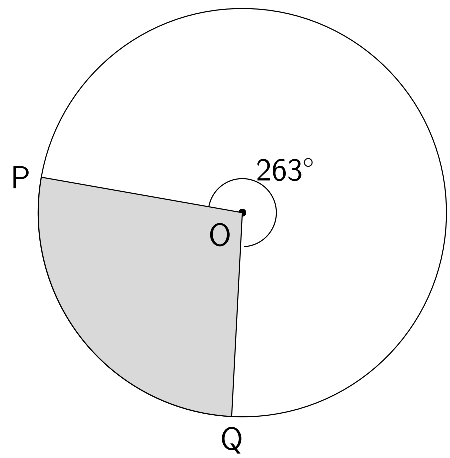
4. A brand of coffee is now sold with a smaller amount of coffee contained in each jar. (3)

   The jars now contain 12.5% less coffee than they previously did, and they now contain 280 grams.

   Calculate how much each jar of coffee used to contain.

5. Determine the gradient of the line with equation $6x + 2y - 10 = 0$. (2)

6. The diagram below shows the circle with centre O, in which the **minor** sector POQ is shaded. (3)



The shaded sector has an area of 94 square centimetres.

Calculate the diameter of the circle.

7. Eight values are ordered from smallest to largest, as shown below.

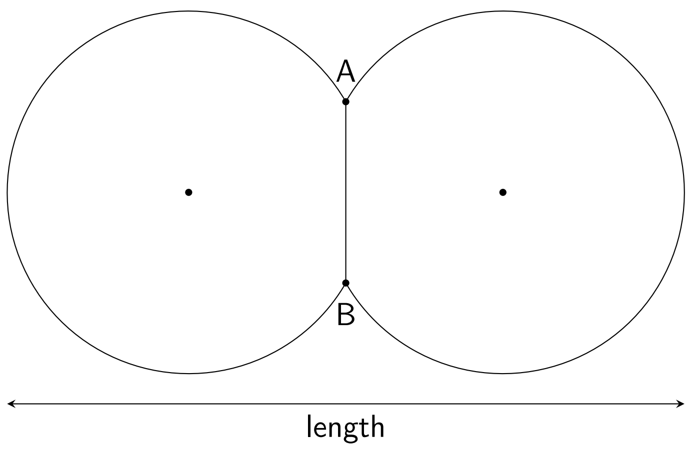
$$-6 \quad -4 \quad 0 \quad 3 \quad k \quad 11 \quad 12 \quad 18$$

(a) Determine the interquartile range of the values. (2)

(b) Given that the median of the values is 4, state the value of k. (1)

8. Solve the equation $5 + \dfrac{12}{x} = 7$. (2)

9. The diagram below shows a shape made of parts of two circles, each with diameter 42 millimetres, which share a common chord, AB. The centre of each circle is marked. (4)



The length of line AB is 21 millimetres. Calculate the length of the shape.

END OF PRACTICE EXAM C PAPER 2

HINTS: Practice Exam C Paper 2

1. Numbers written in scientific notation can be (carefully) typed into a calculator when performing calculations. It **isn't necessary** to write them out in full first...
2. How many **years** is it from 2021 to 2026? What should you **multiply by** to produce a **decrease of 13%**?
3. If the total height is 24cm and the *cylinder* is 17cm, what is the height of the **cone**?
4. What percentage is equivalent to **12.5% less than the original**?
5. Make y the **subject** of the equation...
6. What is the **formula** for the area of a sector? What is the angle within the **minor** sector?
7. Careful when one of the quartiles is **negative**. For part (b), remember that the median is the **middle of the middle values**.
8. To solve an equation containing a fraction, try **multiplying both sides by the denominator**...
9. Isolate a **right-angled triangle** using **half** of line AB and a **radius**.

ANSWERS: Practice Exam C Paper 2

1. 5.9535×10^{24} kilograms
2. £8672.52
3. 4900 cm^3
4. 320 grams
5. -3
6. 21.08 centimetres
7. (a) 13.5
   (b) $k = 5$
8. $x = 6$
9. 78.4 millimetres